

Brownian Motion

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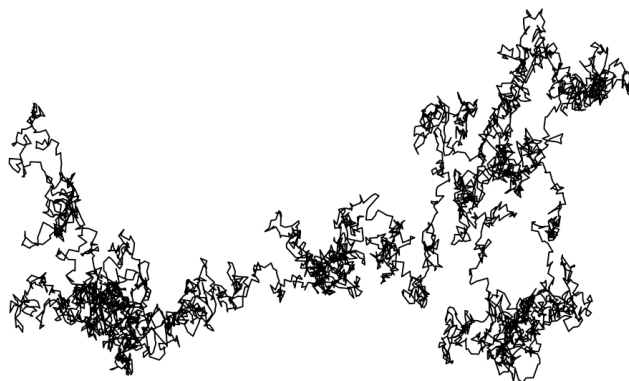
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1 Introduction and Motivation

In 1827, while observing pollen particles suspended in water, botanist Robert Brown noticed that each particle exhibited a continuous, random motion, as illustrated in Figure 1. After repeated experimentation, he determined that these motions arose neither from currents in the fluid, nor from gradual evaporation, but instead belonged to the particles themselves [12, p. 2]. This phenomenon was later named Brownian motion and not much progress was made on the topic until much later. In 1905, Einstein published four revolutionary papers – one on the photoelectric effect, two on special relativity and mass-energy equivalence and, perhaps the least well-known of the four, one on Brownian motion. In this paper he realises that Brownian motion is fundamentally a problem of probability theory [12, p. 5] and the aim of this essay is to develop the necessary mathematical framework to describe it, and to then prove some of its key properties.

Figure 1: A simulation of the path of a pollen particle A.2



In this essay, we will focus on Brownian motion in one dimension, that is the motion of a particle along one co-ordinate axis, although generalising to higher dimensions is relatively natural and straightforward and such a treatment can be found in Resnick [14, § 6]. To fully understand the mathematics of Brownian motion, it is necessary that we begin by developing some preliminary measure theory.

2 Measure Theory

This first chapter is primarily concerned with measuring the ‘length’ (or measure) of subsets of the real numbers. For some subsets, such as intervals or singleton sets, this is fairly intuitive, but our aim is to generalise this to (almost) every subset of $\mathbb{R}$. Developing this idea will later prove to be remarkably useful when developing a rigorous theory of probability over uncountable sets.

2.1 The Borel Sets

We begin by aiming to describe the subsets for which we can assign a length. We will see later that allowing arbitrary subsets of $\mathbb{R}$ is problematic since it is not possible to extend a reasonable notion of measure to every such subset. Hence, it is necessary to characterise a general group of sets for which we can do so, before then defining a general function that assigns a measure to these sets.

Definition 2.1. [6, p. 57] A σ -algebra $\mathcal{F}$ on some set Ω is a collection of subsets of Ω such that $\Omega \in \mathcal{F}$ and $\mathcal{F}$ is closed under complements and countable unions.

By De Morgan's laws, it is easy to show that a σ -algebra is closed under countable intersections as well, and proof of this fact is given in ST120.

Definition 2.2. [6, p. 63] Let Ω be a set and let $\mathcal{F}$ be a σ -algebra on Ω . A *measure* on Ω is a function $\mu : \mathcal{F} \rightarrow [0, \infty]$ such that

1. $\mu(\emptyset) = 0$
2. $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$ for any disjoint subsets $A_i \in \mathcal{F}$.

We say that a measure μ is *finite* if $\mu(\Omega) < \infty$ and that μ is a *probability measure* if $\mu(\Omega) = 1$. We call the ordered triple $(\Omega, \mathcal{F}, \mu)$ a *measure space* and the ordered triple $(\Omega, \mathcal{F}, \mathbb{P})$ a *probability space* in the case that $\mathbb{P}$ is a probability measure.

Lemma 2.3. For any collection $\mathcal{G}$ of subsets of some set Ω , there exists a smallest σ -algebra on Ω that contains $\mathcal{G}$. We denote this sigma-algebra $\sigma(\mathcal{G})$ and call it the σ -algebra generated by $\mathcal{G}$.

Proof. [4, p. 4] The first step of this proof is to show that, in general, the intersection of arbitrary σ -algebras $\{\Sigma_\alpha\}_{\alpha \in \mathcal{A}}$ on a set S is also a σ -algebra Σ on S . To do this, we first note that each Σ_α is a σ -algebra so $S \in \Sigma_\alpha$ for all $\alpha \in \mathcal{A}$ and hence $S \in \Sigma$. Then, if $X \in \Sigma$, then $X \in \Sigma_\alpha$ for every $\alpha \in \mathcal{A}$ so, by closure under complements, $X^c \in \Sigma_\alpha$, for every $\alpha \in \mathcal{A}$, so $X^c \in \Sigma$. Finally, if $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of sets in Σ , then each X_n is in every Σ_α , so $\bigcup_{n \in \mathbb{N}} X_n \in \Sigma_\alpha$ for every $\alpha \in \mathcal{A}$ so $\bigcup_{n \in \mathbb{N}} X_n \in \Sigma$. So, Σ contains S and is closed under complements and countable unions, so is a σ -algebra.

We are now able to prove the existence of smallest σ -algebra containing $\mathcal{G}$. Let $\mathcal{A}$ be some index set and $\mathbb{F} = \{\mathcal{F}_\alpha\}_{\alpha \in \mathcal{A}}$ be the complete collection of σ -algebras on Ω such that $\mathcal{G} \subseteq \mathcal{F}_\alpha$ for every $\alpha \in \mathcal{A}$. Note that $\mathbb{F}$ is non-empty since $\mathcal{P}(\Omega) \in \mathbb{F}$. Hence, setting $\mathcal{F} = \bigcap_{\alpha \in \mathcal{A}} \mathcal{F}_\alpha$, we have that $\mathcal{F}$ is a σ -algebra and then noting that any σ -algebra containing $\mathcal{G}$ also contains $\mathcal{F}$ completes the proof. $\square$

Definition 2.4. [4, p. 5] Let $(X, \mathcal{T})$ be a topological space. The *Borel σ -algebra*, denoted $\mathcal{B}(X)$, is the σ -algebra generated by the open sets in $\mathcal{T}$. That is, $\mathcal{B}(X)$ is the smallest σ -algebra containing $\mathcal{T}$.

For the purposes of this essay we will, unless otherwise specified, restrict ourselves to the set of real numbers equipped with the usual topology, with $\mathcal{B}(\mathbb{R})$ being the smallest σ -algebra containing all open subsets of $\mathbb{R}$. When we say ‘the Borel σ -algebra’ we will be referring to $\mathcal{B}(\mathbb{R})$ and ‘the Borel sets’ will refer to the elements of $\mathcal{B}(\mathbb{R})$.

Let us briefly check whether some familiar sets are Borel sets [3, pp. 40–42]. Since every open subset of $\mathbb{R}$ is in $\mathcal{B}(\mathbb{R})$ and every closed subset of $\mathbb{R}$ has an open complement, we have, by closure under complements, that every closed subset of $\mathbb{R}$ is a Borel set. Then, every singleton set $\{x\}$ is in $\mathcal{B}(\mathbb{R})$ since $\{x\} = [x, x]$ which is closed. It then follows that every finite set is a Borel set, since it can be written as a finite union of singletons and, by closure under countable unions, we also have that every countable set is a Borel set. For half-open intervals, since we can write $[a, b) = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b)$ and $(a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n})$, it follows by closure under countable intersections that half-open intervals are Borel sets.

2.2 The Outer Measure

We now aim to develop a notion of measure for every Borel set beginning with intervals on $\mathbb{R}$. Consider the extended real line that includes a point at infinity and a point at negative infinity denoted $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$. Let S denote the set of all intervals on $\mathbb{R}$ and define a function $\ell : S \rightarrow \overline{\mathbb{R}}$ that defines the length of an interval in the intuitive way. That is, for any bounded interval $I \in S$ with endpoints a, b and $a \leq b$, we define $\ell(I) = b - a$ and for any half-bounded or unbounded interval, $I = (-\infty, b)$, $(-\infty, b]$, (a, ∞) , $[a, \infty)$ or $(-\infty, \infty)$, we say that $\ell(I) = \infty$. We remark that for a singleton set $\{a\}$, the length is given by $\ell(\{a\}) = \ell([a, a]) = 0$ and, since the empty set $\emptyset = (0, 0)$, it follows that $\ell(\emptyset) = 0$. We can now generalise to every subset of $\mathbb{R}$.

Definition 2.5. [1, p. 14] The (*Lebesgue*) *outer measure* of a set $A \subseteq \mathbb{R}$ is given by

$$m^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n, \text{ each } I_n \text{ an open interval} \right\}.$$

That is, if X is the collection of every set of open intervals $I_n \subseteq \mathbb{R}$ such that A is contained within the union of these intervals, and we sum the lengths of each interval for every element of X , then the smallest such sum is the outer measure of A . It is reasonable at this point to check that this definition satisfies the properties we would expect from a length-defining function. Certainly, we would hope that $m^*(I) = \ell(I)$ for any interval $I \in S$. Also if $A \subseteq B \subseteq \mathbb{R}$ then we would like to have that $m^*(A) \leq m^*(B)$ and if $(A_i)_{i \geq 1}$ is a sequence of subsets of $\mathbb{R}$, then

$$m^* \left(\bigcup_{i=1}^{\infty} A_i \right) \leq \sum_{i=1}^{\infty} m^*(A_i).$$

Luckily, it turns out that each of these properties is in fact true [3, pp. 22–25] and the outer measure is a measure, but their proofs are not strictly relevant for the purposes of

this essay and have thus been omitted. We will however give an elegant proof of another (at first glance, perhaps slightly more surprising) property in order to motivate our studies of probability theory over uncountable sets.

Lemma 2.6. *Every countable subset of $\mathbb{R}$ has outer measure 0.*

Proof. [1, pp. 15-16] Let $A = \{x_1, x_2, \dots\}$ be a countable subset of $\mathbb{R}$ and set $\varepsilon > 0$. For $n \in \mathbb{N}$, let

$$I_n = \left(x_n - \frac{\varepsilon}{2^n}, x_n + \frac{\varepsilon}{2^n}\right).$$

Then $l(I_n) = \frac{\varepsilon}{2^{n-1}}$. As $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 2$, it follows that $\sum_{i=1}^{\infty} l(I_n) = 2\varepsilon$ so $m^*(A) = 0$. $\square$

In addition to the above properties, there is one further property that we would like to have, namely countable additivity. That is, for disjoint subsets A and B , $m^*(A \cup B) = m^*(A) + m^*(B)$. However, in general, this does not hold and it is in fact possible to construct a subset of $\mathbb{R}$ that is not measurable, Vitali's non-measurable set [10, § 1] being a nice example. This motivates the following definition.

Definition 2.7. [3, p. 27] A subset $E \subseteq \mathbb{R}$ is (*Lebesgue*) *measurable* if

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$$

for every $A \subseteq \mathbb{R}$.

It turns out that selecting precisely the subsets $E \subseteq \mathbb{R}$ that satisfy the equation in definition 2.7 will guarantee that each property we expect of a measure is satisfied. Note that, in the above definition, we say that $a + \infty = \infty + a = \infty + \infty = \infty$. Unless otherwise specified, we will denote the collection of Lebesgue measurable subsets of $\mathbb{R}$ by $\mathcal{M}$ and, for each $A \in \mathcal{M}$, we will denote the measure of A by $m(A)$.

It also turns out that the collection $\mathcal{M}$ of Lebesgue measurable subsets of $\mathbb{R}$ forms a σ -algebra [3, p. 29-34] however proof of this property is rather long and not strictly relevant since we are more interested in the collection of Borel sets. This leads us on to the following proposition.

Proposition 2.8. Every Borel set is Lebesgue measurable.

Proof. [3, p. 42] We know that $\mathcal{M}$ is a σ -algebra and every open interval $(a, b) \in \mathcal{M}$. Also, $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open intervals, so it follows that $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}$ and hence every Borel set is Lebesgue measurable. $\square$

Example 2.9. Consider the following question from [3, p. 12]. If a number is chosen uniformly at random from the interval $[0, 1]$, what is the probability that this number is rational?

We have now built up enough theory to give a reasonable answer to this question. Consider the measure space $([0, 1], \mathcal{B}([0, 1]), m)$, where m is the Lebesgue measure restricted to $[0, 1]$. Then, since $[0, 1] \cap \mathbb{Q}$ is countable, it follows that $m([0, 1] \cap \mathbb{Q}) = 0$, and since $[0, 1] \cap \mathbb{R} \setminus \mathbb{Q}$ and $[0, 1] \cap \mathbb{Q}$ are disjoint, we have that $m([0, 1] \cap \mathbb{R} \setminus \mathbb{Q}) = 1$. So,

defining the probability of choosing an element from a Borel subset of $[0, 1]$ to be equivalent to the Lebesgue measure of that subset gives us that the probability of choosing a rational in $[0, 1]$ is 0, whilst the probability of choosing an irrational is 1.

Example 2.10. We also give another interesting example from Folland [6, p. 38]. Recall the Cantor set C from MA260. That is, the set constructed by, beginning with $[0, 1]$, iteratively removing the middle third of each remaining interval. Since we first remove $1/3$ of an interval, then $2/9$, and then $4/27$ and so on in this way, we have that

$$m(C) = 1 - \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = 1 - \frac{1}{3} \times \frac{1}{1 - (2/3)} = 1 - 1 = 0.$$

Hence, the Cantor set is an example of a set that has the cardinality of the continuum, yet measure zero.

In general, an event $E \in \mathcal{F}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is said to happen *almost surely* if $\mathbb{P}(E^c) = 0$ so, in the first example, an irrational number will be chosen almost surely and in the second, an element of $[0, 1] \setminus C$ will be chosen almost surely.

2.3 Functions and Integration

Definition 2.11. [5, p. 94] Let $(\Omega, \mathcal{F}, m)$ be a measure space. Then, a function $f : \Omega \rightarrow \mathbb{R}$ is *measurable* if

$$f^{-1}(B) = \{x \in \Omega : f(x) \in B\} \in \mathcal{F}$$

for every Borel set $B \in \mathcal{B}(\mathbb{R})$. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, then a measurable function on Ω is called a *random variable*.

Before we give an example to illustrate the idea of a measurable function we first recall the definition of an indicator function. Let $(\Omega, \mathcal{F}, m)$ be a measure space. Then, the *indicator function* for some subset $A \subseteq \Omega$ is a function $\mathbf{1}_A : \Omega \rightarrow \mathbb{R}$ given by

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A; \\ 0, & \text{otherwise.} \end{cases}$$

Example 2.12. Adapted from Capiński and Kopp [3, p. 59]. Let $([0, 1], \mathcal{B}([0, 1]), m)$ be a measure space, where m is the Lebesgue measure restricted to $[0, 1]$. We now define a set $Q := [0, 1] \cap \mathbb{Q}$ and consider whether or not the indicator function $\mathbf{1}_Q : [0, 1] \rightarrow \mathbb{R}$ given by

$$\mathbf{1}_Q(x) = \begin{cases} 1, & \text{if } x \in Q; \\ 0, & \text{otherwise,} \end{cases}$$

is measurable. To do this, take some Borel set $B \in \mathcal{B}(\mathbb{R})$. If $0 \in B$ and $1 \in B$, then $\mathbf{1}_Q^{-1}(B) = [0, 1]$; if $0 \in B$ and $1 \notin B$, then $\mathbf{1}_Q^{-1}(B) = [0, 1] \setminus \mathbb{Q}$; if $0 \notin B$ and $1 \in B$, then $\mathbf{1}_Q^{-1}(B) = [0, 1] \cap \mathbb{Q}$ and if $0 \notin B$ and $1 \notin B$, then $\mathbf{1}_Q^{-1}(B) = \emptyset$. So, since $\mathbf{1}_Q^{-1}(B) \in \mathcal{B}([0, 1])$ in any of these cases, it follows that $\mathbf{1}_Q$ is a measurable function.

Using the idea of measure, we now have the tools to construct a far more powerful version of the integral than the Riemann integral. Consider the indicator function $\mathbf{1}_Q$ on the set $Q := [0, 1] \cap \mathbb{Q}$. We know that the rationals are dense in the reals, so any non-degenerate interval $I \subseteq [0, 1]$ contains both rationals and irrationals. Hence, the upper integral of $\mathbf{1}_Q$ is equal to 1, while the lower integral is 0, so $\mathbf{1}_Q$ is not Riemann integrable.

We should however ask ourselves what we expect the value of this integral to be. Given there are countably many points over which the value of this function deviates from 0, it would be reasonable to expect that, since the measure of a countable set is 0 and the measure of $[0, 1]$ is 1, the integral of $\mathbf{1}_Q$ between 0 and 1 should be 0. This motivates the following idea from Capiński and Kopp [3, pp. 75–78].

We will denote the integral for some function f on a measure space $(\Omega, \mathcal{F}, m)$ by $\int_{\Omega} f dm$. Generalising the idea above, for some measure space $(\Omega, \mathcal{F}, m)$, and some $A \in \mathcal{F}$ we would like that

$$\int_{\Omega} \mathbf{1}_A dm = m(A).$$

Our approach for developing this integral will be to progressively incorporate a broader range of functions on which the integral is defined. So far, we have a reasonable notion of the integral for indicator functions, so let us now define the next class of function on which we can define the integral.

Definition 2.13. [3, pp. 75–76] A *simple function* is a function that can be written in the form

$$\sum_{i=1}^n c_i \mathbf{1}_{A_i}$$

where A_i is a measurable subset of $\mathbb{R}$ for all $i = 1, \dots, n$. Then, for $c_i \geq 0$, we define the integral of a simple function to be

$$\int_{\Omega} \sum_{i=1}^n c_i \mathbf{1}_{A_i} dm = \sum_{i=1}^n c_i m(A_i).$$

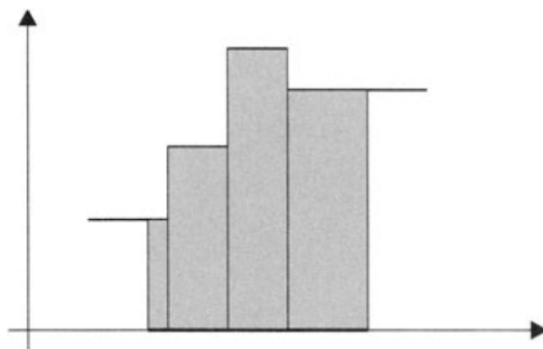
as illustrated in Figure 2. Note that the heights of each rectangle in Figure 2 correspond to each c_i and the widths of each rectangle correspond to the measure of each A_i .

Note that, without the restriction that each c_i be greater than 0, this definition would give us problems. Suppose that there exist $i, j \in \{1, \dots, n\}$ such that $m(A_i) = m(A_j) = \infty$. Then, without loss of generality, if $c_i > 0$ and $c_j < 0$, then we end up with the sum $\infty - \infty$ which is not well-defined. To get around this issue, we could restrict the subsets A_i to those with finite measure, however it is often convenient to include subsets of infinite measure, so instead we take each $c_i \geq 0$. We then adopt the convention that $\infty + \infty = \infty$ and $\infty + a = a + \infty = \infty$ for every $a \in \mathbb{R}$.

We can now use this definition to define the Lebesgue integral for non-negative measurable functions. Let $f : \mathbb{R} \rightarrow [0, \infty]$ be a measurable function. Then we define

$$\int_{\mathbb{R}} f dm = \sup \left\{ \int_{\mathbb{R}} \phi : \phi \text{ is simple and } 0 \leq \phi \leq f \right\}.$$

Figure 2: Integral of a simple function [3, p. 76]



That is, for any measurable function $f : \mathbb{R} \rightarrow [0, \infty]$, we can get an arbitrarily good approximation for the integral of this function by integrating simple functions below f .

It is now easy to extend this definition to arbitrary measurable functions. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function. We first decompose f into its positive and negative parts. That is, let $f^+ := \max(f, 0)$ and $f^- := \max(-f, 0)$. We then define the integral of f to be

$$\int_{\mathbb{R}} f dm = \int_{\mathbb{R}} f^+ dm - \int_{\mathbb{R}} f^- dm.$$

It is indeed possible for the integral of a function to be infinite under this definition, possibly the simplest being the constant function $\mathbf{1}_{\mathbb{R}} : \mathbb{R} \rightarrow \mathbb{R}$, $\mathbf{1}_{\mathbb{R}}(x) = 1$ for every $x \in \mathbb{R}$, which we denote $\mathbf{1}_{\mathbb{R}}$ since it can be thought of as the indicator function for $\mathbb{R}$. Therefore, we say that a function f is integrable if $\int_{\mathbb{R}} |f| dm < \infty$.

Definition 2.14. [3, p. 114] If $X : \Omega \rightarrow \mathbb{R}$ is a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the *expectation* of X is denoted $\mathbb{E}[X]$ and is given by

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}.$$

We finish this section with a result about expectation that will be useful later.

Lemma 2.15. *Let $X : \Omega \rightarrow \mathbb{R}$ be a non-negative random variable. Then, X is almost surely 0 if and only if $\mathbb{E}[X] = 0$.*

Proof. Adapted from Folland [6, pp. 110–111]. We first prove the reverse direction. Suppose for contradiction that $\mathbb{E}[X] = 0$ and the set $A = \{\omega \in \Omega : X(\omega) > 0\}$ has strictly positive measure. That is, $\mathbb{P}(A) > 0$. Since X is non-negative and $A \subseteq \Omega$, we have that the integral of X on Ω is at least the integral of X on A , so

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P} \geq \int_A X d\mathbb{P} > 0,$$

contradicting the fact that $\mathbb{E}[X] = 0$. Therefore, we must have that $\mathbb{P}(A) = 0$. To prove the forward direction, we first note that this result is fairly trivial when $X : \Omega \rightarrow \mathbb{R}$

is a simple function since, for subsets $A_i \subseteq \Omega$, $i \in \{1, \dots, n\}$, we have that $X(\omega) = \sum_{i=1}^n c_i \mathbf{1}_{A_i}(\omega)$ and thus, $\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P} = 0$ if, and only if, for each $i = 1, \dots, n$, either $a_i = 0$ or $\mathbb{P}(A_i) = 0$. We now proceed with the forward direction for general non-negative random variables. Suppose that $X(\omega) = 0$ almost surely. Then, if $Y : \Omega \rightarrow \mathbb{R}$ is a simple measurable function with $0 \leq Y(\omega) \leq X(\omega)$ for every $\omega \in \Omega$, we have that $Y(\omega) = 0$ almost surely. Hence, for Y simple, we have that

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P} = \sup_{Y \leq X} \int_{\Omega} Y d\mathbb{P} = 0,$$

as required. $\square$

3 Brownian Motion

Our attention will now be turned towards applying the concepts we have already developed to formally prove various properties of Brownian motion. We begin by recalling the definition of a stochastic process from ST227.

Definition 3.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and T an index set, typically interpreted as representing a set of times. A *stochastic process* is a collection of random variables $\{X_t : t \in T\}$ such that each random variable X_t is defined on Ω and takes values in a set S , called the *state space*.

Alternatively, we can regard a stochastic process as a function $X : T \times \Omega \rightarrow S$ such that for each fixed $t \in T$, the mapping $X_t : \Omega \rightarrow S$, $X_t(\omega) := X(t, \omega)$ is a random variable [11, p. 2]. We can then fix an outcome $\omega \in \Omega$, rather than fixing $t \in T$, so that the function $t \mapsto X(t, \omega) : T \rightarrow S$ describes what we call a *sample path* of the process, which is a deterministic function of t . We remark that, for any stochastic process, when we write X_t , we will be referring to the random variable $X_t : \Omega \rightarrow S$ and when we write $X(t, \omega)$, we will be referring to the value of the function X at the point $(t, \omega) \in T \times \Omega$. We are now ready to present the main definition of the essay, adapted from Mörters and Peres [13, p. 7] and Billingsley [2, p. 498].

Definition 3.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Then, a stochastic process $\{B_t : t \in [0, \infty)\}$, with state space $\mathbb{R}$, is called a *Brownian motion (on Ω)* if it satisfies the following properties.

1. Almost surely, the process starts at some $x \in \mathbb{R}$. Specifically, $\mathbb{P}(B_0 = x) = 1$.
2. Almost surely, the sample paths of the process are continuous. That is,

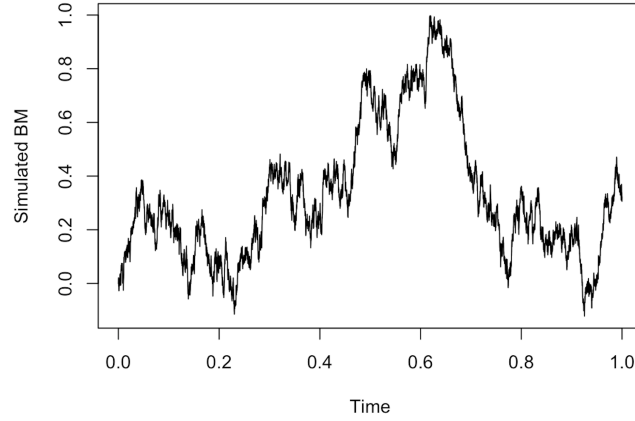
$$\mathbb{P}(\omega \in \Omega : t \mapsto B(t, \omega) \text{ is continuous on } [0, \infty)) = 1.$$

3. The process has independent increments. In particular, for any collection of times $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$, the increments $B_{t_2} - B_{t_1}, B_{t_3} - B_{t_2}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent random variables.

4. For all $t \geq 0$ and $h > 0$, the increment $B_{t+h} - B_t$ is normally distributed with expectation zero and variance h .

That is, a Brownian motion is just an uncountable collection of random variables $\omega \mapsto B(t, \omega)$, on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$, satisfying each of the above properties. Note that, in the case of $x = 0$ in the first property, we say that B is a *standard Brownian motion* and, for each fixed $\omega \in \Omega$, we say that $t \mapsto B(t, \omega)$ is a *standard sample path*, as in figure 3.

Figure 3: A standard sample path in one-dimension A.2



It is important to remark at this stage that these are the properties that we would like Brownian motion to have based on observations, and in no way guarantees the existence of such a process. So, is it possible to construct a stochastic process satisfying all four of these properties? Well, it turns out that it is in fact possible [14, pp. 489–494], but such a construction is fairly convoluted. For the purpose of this essay, we work under the assumption that Brownian motion does indeed exist as we are primarily interested in the further properties it has.

3.1 Basic Properties and Scaling

Let us begin by finding the expectation of a standard Brownian motion. More precisely, we will aim to find the expectation of each random variable B_t for each fixed $t \in [0, \infty)$. Noting that linearity still holds for the Lebesgue integral, we have that $\mathbb{E}[B_t] = \mathbb{E}[B_0] + \mathbb{E}[B_t - B_0]$. Then, by normally distributed increments with expectation 0, we have that $\mathbb{E}[B_t - B_0] = 0$ and since $B_0 = 0$ almost surely, we have that $B_0 = 0$ everywhere except some set of measure zero $A \subseteq \Omega$, so

$$\mathbb{E}[B_0] = \int_{\Omega} B_0 d\mathbb{P} = \int_{\Omega \setminus A} 0 d\mathbb{P} + \int_A B_0 d\mathbb{P} = 0 + 0 = 0.$$

Hence, $\mathbb{E}[B_t] = 0$ for every $t \in [0, \infty)$. This should make sense intuitively as a standard Brownian motion almost surely starts at 0 and, since it is a random process, there is no

deterministic tendency for it to move in any specific direction.

We now present the following scaling theorems, noting that their proofs are merely exercises in the definition of Brownian motion and have thus been relegated to appendix A.1.

Proposition 3.3 (Scaling invariance). [13, p. 12] Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{B_t : t \geq 0\}$ be a standard Brownian motion on Ω . Then, the stochastic process $\{X_t : t \geq 0\}$ given by $X_t = \frac{1}{a}B_{a^2t}$, for all $t \in [0, \infty)$, is also a standard Brownian motion on Ω .

Proposition 3.4 (Time inversion). [13, p. 13] Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{B_t : t \geq 0\}$ be a standard Brownian motion on Ω . Then, the stochastic process $\{X_t : t \geq 0\}$ given by

$$X_t = \begin{cases} 0, & \text{for } t = 0, \\ tB_{1/t}, & \text{for } t > 0, \end{cases}$$

is also a standard Brownian motion on Ω .

Corollary 3.5 (Law of Large Numbers). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{B_t : t \geq 0\}$ be a standard Brownian motion on Ω . Then, $\lim_{t \rightarrow \infty} B_t/t = 0$, almost surely.*

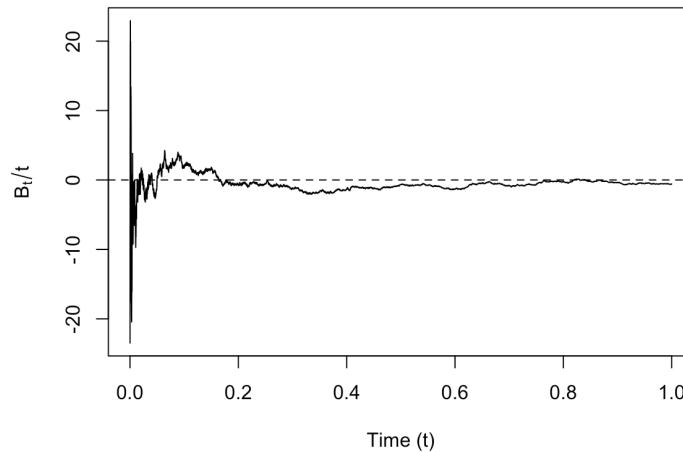
Proof. [13, p. 14] This follows from proposition 3.4 as

$$\lim_{t \rightarrow \infty} \frac{B_t(\omega)}{t} = \lim_{t \rightarrow \infty} \frac{B(t, \omega)}{t} = \lim_{t \rightarrow \infty} X\left(\frac{1}{t}, \omega\right) = X(0, \omega) = X_0(\omega),$$

and, since X_t is a standard Brownian motion, we have that $X_0(\omega) = 0$, almost surely. $\square$

That is, the deterministic function $t \mapsto t$ almost surely gets large faster than the random function $t \mapsto B(t, \omega)$. This phenomenon is illustrated in Figure 4.

Figure 4: The law of large numbers for Brownian motion A.2



3.2 The Fractal Paths of Brownian Motion

An interesting, and perhaps slightly surprising, property of the sample paths of Brownian motion is that, almost surely, they are nowhere differentiable. This tells us that each sample path has a fractal-like structure, in that it exhibits ‘roughness’ regardless of how far you zoom in yet, at the same time, it is somewhat regular in the sense that it is continuous almost surely. Proof of this property [13, p. 21] is rather convoluted and is thus omitted from this essay, but we do present a proof of the following slightly weaker proposition.

Proposition 3.6 (Monotonicity). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{B_t : t \geq 0\}$ be a Brownian motion on Ω . Then, for all $a, b \in [0, \infty)$ with $a < b$ and all $\omega \in \Omega$, the sample path $t \mapsto B(t, \omega)$ is almost surely not monotonic on $[a, b]$.

Proof. [13, p. 18] Fix $\omega \in \Omega$ and take some interval $[p, q] \subset [0, \infty)$ with $p, q \in \mathbb{Q}$ and $p < q$. Now, let $\{p = t_0, t_1, t_2, \dots, q = t_n\}$ be a partition of $[p, q]$, with each $t_i \in \mathbb{Q}$. Then, by independence of increments, the probability that each $B(t_{i+1}, \omega) - B(t_i, \omega)$ is positive is 2^{-n} and likewise, that each is negative is also 2^{-n} . Hence, the probability that each $B(t_{i+1}, \omega) - B(t_i, \omega)$ has the same sign is $2^{-n} + 2^{-n} = 2 \cdot 2^{-n} = 2^{1-n}$, which approaches 0 as $n \rightarrow \infty$. Therefore, the probability that $t \mapsto B(t, \omega)$ is monotonic on $[p, q]$ is 0. We now need to consider all intervals with rational endpoints simultaneously. We know that the set of all closed and bounded intervals $[p_i, q_i]$, with $p_i, q_i \in \mathbb{Q}$, is countable so, by Boole’s inequality, we have that

$$\mathbb{P} \left(\bigcup_{i=1}^{\infty} t \mapsto B(t, \omega) \text{ monotonic on } [p_i, q_i] \right) \leq \sum_{i=1}^{\infty} \mathbb{P} (t \mapsto B(t, \omega) \text{ monotonic on } [p_i, q_i]) = 0$$

so, almost surely, $t \mapsto B(t, \omega)$ is not monotonic on every interval with rational endpoints. Then, since $\mathbb{Q}$ is dense in $\mathbb{R}$, every interval $[a, b]$ with $a < b$ will have some sub-interval $[p_i, q_i]$ with $p_i < q_i$ and $p_i, q_i \in \mathbb{Q}$. Hence, if $t \mapsto B(t, \omega)$ were monotonic on any interval $[a, b]$, it would necessarily be monotonic on a sub-interval $[p_i, q_i]$, which we have proven to be almost surely impossible. Thus, Brownian sample paths are almost surely not monotonic on every interval $[a, b]$. $\square$

We should remark that this argument only works since the collection of sub-intervals of $[p, q]$ with rational end-points is countable. That is, it is not sufficient to simply choose a general interval $[a, b] \subset \mathbb{R}$ and take a countable sequence of smaller partitions as there are uncountably many partitions of $[a, b]$.

3.3 The Markov Property

In the context of stochastic processes, the Markov property refers to the notion of ‘memorylessness’. That is, a Markov process depends only on a single previous state at some time t , independent of all states before t . The aim of this section is to prove that Brownian motion is indeed a Markov process and to then strengthen this notion slightly in preparation for the final section of the essay. We begin with the following idea.

We say that any two stochastic processes $\{X_t : t \in T\}$ and $\{Y_t : t \in T\}$ are *independent* if, for any times $t_1, \dots, t_n \geq 0$ and $s_1, \dots, s_m \geq 0$, the random vectors $(X_{t_1}, \dots, X_{t_n})$ and $(Y_{s_1}, \dots, Y_{s_m})$ are independent [13, p. 37]. That is, each pair of random variables X_{t_i} and Y_{t_j} are independent. We can now prove our first version of the Markov property.

Proposition 3.7 (The Markov Property). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\{B_t : t \geq 0\}$ be a standard Brownian motion on Ω and fix $s > 0$. Then, the stochastic process $\{B_{t+s} - B_s : t \geq 0\}$ is also a standard Brownian motion on Ω , independent of $\{B_t : 0 \leq t \leq s\}$.

Proof. Developed from Mörters and Peres [13, p. 37]. Fix $\omega \in \Omega$. At $t = 0$, we have that $B_{t+s} - B_s = B_s - B_s$ which is almost surely equal to 0, so the process almost surely starts at 0. Then, for path continuity, $t \mapsto B(t+s, \omega) - B(s, \omega)$ is continuous since $B(s, \omega)$ is constant and $t \mapsto B(t+s, \omega)$ is just a translation of the continuous function $t \mapsto B(t, \omega)$. Next, for any $0 \leq t_1 < t_2 < \dots < t_n$, we have $(B_{t_{i+1}+s} - B_s) - (B_{t_i+s} - B_s) = B_{t_{i+1}+s} - B_{t_i+s}$ which is independent of each other increment $B_{t_{k+1}+s} - B_{t_k+s}$, where $k \neq i$, by the independent increment property of Brownian motion. This increment is also normally distributed with mean 0 and variance $t_i - t_j$ also from the definition of Brownian motion. Hence, $\{B_{t+s} - B_s : t \geq 0\}$ is a standard Brownian motion on Ω . Finally, independence follows since $B_{t+s} - B_s$ is independent of $B_t - B_0 = B_t$ by independent increments. $\square$

We therefore have that Brownian motion is indeed a Markov process. However, it is possible to strengthen this property and we will do so in preparation for the final section of this essay. We remark that the remainder of this section on the Markov property serves to provide an intuitive understanding of the strong Markov property and thus, much of the rigour will be omitted. A more rigorous, yet far more convoluted, treatment of this topic can be found in Mörters and Peres [13, pp. 37–44]. We begin by developing some intuition behind what is meant by a *filtration*.

A filtration is a collection of σ -algebras, $\{\mathcal{F}_t\}_{t \in T}$ such that each $\mathcal{F}_t$ contains all information available about a process up to the time t . To illustrate why this is a useful idea to consider in the context of stochastic processes, let us consider the following simple example.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and consider the stochastic process that arises from repeatedly flipping a fair coin with outcomes heads (**H**) and tails (**T**). Then $T = \mathbb{N}$ and Ω is the set of all infinite sequences of heads and tails. We can then consider a filtration $\mathbb{F} = \{\mathcal{F}_n\}_{n \in \mathbb{N}}$ that represents all events that could have occurred in this stochastic process up to some time t . If we let $\{\mathbf{H}\}$ represent all sequences with the first flip being a head, $\{\mathbf{TH}\}$ represents all sequences with the first flip being a tail and the second being a head, and likewise for all other sequences in this way, then $\mathcal{F}_1 = \{\emptyset, \{\mathbf{H}\}, \{\mathbf{T}\}, \Omega\}$ and $\mathcal{F}_2$ is the σ -algebra generated by $\{\{\mathbf{HH}\}, \{\mathbf{HT}\}, \{\mathbf{TH}\}, \{\mathbf{TT}\}\}$. This continues for each $n \in \mathbb{N}$ where each successive $\mathcal{F}_n$ contains all information about the process available at time n . Notice that $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ for each $n \in \mathbb{N}$ since all information about the process available at time n is certainly also available at time

$n + 1$. We also remark that $\mathcal{F}_0 = \{\emptyset, \Omega\}$ since, at time $t = 0$, we don't know anything about the sequence of flips.

In the context of Brownian motion we usually consider one of the following two filtrations. First, the filtration $\{\mathcal{F}_t^0\}_{t \geq 0}$ representing all information available about the Brownian motion up to time t , or second, the filtration $\{\mathcal{F}_s^+\}_{s \geq 0}$ defined by

$$\mathcal{F}_s^+ = \bigcap_{t > s} \mathcal{F}_t^0$$

for each $s \geq 0$. That is, $\mathcal{F}_s^+$ contains all information in $\{\mathcal{F}_t^0\}_{t \geq 0}$ but also includes an ‘infinitesimal glance into the future’ [13, p. 37–38]. We are now ready to present our final definition before the strong Markov property.

Definition 3.8. [13, p. 40] Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_t\}_{t \in T}$, where T is a well-ordered index set. Then, a random variable $\tau : \Omega \rightarrow T \cup \{\infty\}$ is called a *stopping time* if

$$\{\omega \in \Omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$$

for every $t \in T$.

That is, a stopping time is a random time τ that depends on information up to that time but not beyond that time. In our coin flip example, we have that ‘the time at which the second tail is flipped’ is a stopping time but ‘the last head before the third tail is flipped’ is not a stopping time since we do not yet know the value of the next coin flip.

We are now able to quote the strong version of the Markov property. Note that we will not worry about precisely what it means for a stochastic process to be independent of a filtration in the following proposition, but can intuitively think about this as the process being independent of any information available in that filtration. Proof of the strong Markov property is beyond the scope of this essay and can be found in Mörters and Peres [13, p. 43–44].

Proposition 3.9 (The Strong Markov Property). [13, p. 43] For every almost surely finite stopping time τ , the process

$$\{B(\tau + t) - B(\tau) : t \geq 0\}$$

is a standard Brownian motion independent of $\mathcal{F}^+(t)$.

3.4 The Zero Set

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{B_t : t \geq 0\}$ be a standard Brownian motion on Ω . Then, for each sample path $t \mapsto B(t, \omega)$, the zero set of that sample path is defined as $\mathcal{Z}_\omega := \{t \geq 0 : B(t, \omega) = 0\}$. That is, for each $\omega \in \Omega$, $\mathcal{Z}_\omega$ is the set of times $t \in [0, \infty)$ such that the sample path $t \mapsto B(t, \omega)$ returns to 0. Is it clear by the first property of Brownian motion that, almost surely, $0 \in \mathcal{Z}_\omega$ but there are many questions that we could ask about this set that may not be immediately clear. For instance, is this set

finite, countable or uncountable? Is it bounded? Is it compact? This final section of the essay will aim to prove some nice measure-theoretic and topological properties of $\mathcal{Z}_\omega$ in order to answer some of these questions. We begin with the following proposition.

Proposition 3.10. For each $\omega \in \Omega$, $\mathcal{Z}_\omega$ almost surely has Lebesgue measure zero.

Proof. Adapted from Karatzas and Shreve [9, p. 105]. Consider the random variable $m(\mathcal{Z}_\omega)$, where m denotes the Lebesgue measure. By lemma 2.15, since the Lebesgue measure is non-negative, it suffices to show that $\mathbb{E}[m(\mathcal{Z}_\omega)] = 0$. Noting that Fubini's theorem from MA263 still holds for the Lebesgue integral and allows us to interchange the order of double integrals [3, p. 171], we have that

$$\mathbb{E}[m(\mathcal{Z}_\omega)] = \mathbb{E} \left[\int_0^\infty \mathbf{1}_{\{B_t=0\}} dt \right] = \int_0^\infty \mathbb{E}[\mathbf{1}_{\{B_t=0\}}] dt = \int_0^\infty \mathbb{P}(B_t = 0) dt.$$

Then, since B_t is a random variable with state space $\mathbb{R}$ and $\{0\} \in \mathbb{R}$ has measure zero, $\mathbb{P}(B_t = 0) = 0$ for every $t \in [0, \infty)$. It follows that $\int_0^\infty \mathbb{P}(B_t(\omega) = 0) dt = 0$ and hence, $\mathbb{E}[m(\mathcal{Z}_\omega)] = 0$, as required. $\square$

Therefore, similarly to the set of rationals from example 2.9 and the Cantor set from example 2.10, we have yet another example of a set with Lebesgue measure zero. Our next aim will be to prove that the zero set is almost surely a perfect set. In particular, we would like to show that, for every $\omega \in \Omega$, with probability one, $\mathcal{Z}_\omega$ is closed and has no isolated points. There are two key results that are not proven in this essay that we will use in this proof. The first is that the zero set is almost surely unbounded and the second is that, for any $\varepsilon > 0$, a standard Brownian sample path has infinitely many zeros in the interval $[0, \varepsilon]$ with probability one [9, pp. 104–105]. We now proceed with the proof.

Proposition 3.11. Almost surely, the zero set is closed and has no isolated points.

Proof. [13, p. 48] Fix $\omega \in \Omega$. Closure follows almost immediately from the almost sure continuity of sample paths since for any sequence $(t_n) \in \mathcal{Z}_\omega$ with $t_n \rightarrow t$, we have that $B(t_n, \omega) = 0$ for all n so, by continuity,

$$B(t, \omega) = \lim_{n \rightarrow \infty} B(t_n, \omega) = \lim_{n \rightarrow \infty} 0 = 0,$$

so $t \in \mathcal{Z}_\omega$. We now prove that $\mathcal{Z}_\omega$ has no isolated points. Fix $q \in \mathbb{Q}$ and define $\tau_q := \inf\{t \geq q : B(t, \omega) = 0\}$. Then τ_q is a stopping time and since the probability that the zero set is unbounded is one, τ_q is almost surely finite. By the Strong Markov property, we have that $B_{t+\tau_q} - B_{\tau_q}$ is a standard Brownian motion so, noting that a Brownian sample path has infinitely many zeros in any interval $[0, \varepsilon]$ with $\varepsilon > 0$, it follows that τ_q is not isolated from the right.

We now need to check that τ_q is not isolated from the left. Let $t \in \mathcal{Z}_\omega$ be different to every τ_q with $q \in \mathbb{Q}$ and let (q_n) be an increasing sequence with $q_n \in \mathbb{Q}$ for every n , such that $q_n \rightarrow t$. Then, defining the sequence $t_n = \tau_{q_n}$, we have that $t_n \geq q_n$ since each t_n is in $\{t : t \geq q_n\}$ and that $t_n < t$ since each t_n is the infimum of $\{t : t \geq q_n\}$. Since $q_n \rightarrow t$, we have that $t_n \rightarrow t$, and hence t is not isolated from the left. $\square$

We finish this section with a concluding discussion about the some further properties of $\mathcal{Z}_\omega$. If we define $\mathcal{Z}_\omega^\tau := \mathcal{Z}_\omega \cap [0, \tau]$, that is let $\mathcal{Z}_\omega^\tau$ be the zero set restricted to some bounded time interval $[0, \tau]$, then $\mathcal{Z}_\omega^\tau$ is clearly bounded and is closed by proposition 3.11. Therefore, by the Heine-Borel theorem, $\mathcal{Z}_\omega$ is compact. It can also be shown that $\mathcal{Z}_\omega^\tau$ is metrisable and totally disconnected though we don't concern ourselves with the proofs of these properties. Thus, by Brouwer's theorem [7, pp. 8–9], the zero set of Brownian motion on some finite time interval $[0, \tau]$ is homeomorphic to the Cantor set from example 2.10. It is worth taking a moment to reflect on the significance of this result. With probability one, the zero set of Brownian motion, a natural and widely studied object in the context of particle motion, is homeomorphic to the Cantor set, a seemingly unrelated and abstract construct frequently encountered in real analysis.

4 Conclusion

This essay has provided a rigorous foundation for understanding Brownian motion, beginning with measure theory and progressing to its formal probabilistic definition and some of its key properties such as monotonicity, the Markov property and the structure of its zero set.

One natural progression from this essay is the study of stochastic calculus. Unlike 'ordinary' calculus that focuses on deterministic functions, stochastic calculus focuses on functions that exhibit random fluctuations such as Brownian sample paths. This topic is widely applicable in the field of mathematical finance [8, Preface], for instance, it is essential in the derivation of the Black-Scholes equation, possibly the most famous formula in finance, used for pricing European options. That is, computing a fair price for the right but not the obligation to buy or to sell a financial security, such as a stock, on some agreed upon expiration date. A treatment of this subject can be found in Karatzas and Shreve [9] or Cohen and Elliot [4].

A Appendix

A.1 Proof of Scaling Properties

In this section we present the proofs of the aforementioned scaling properties of Brownian motion in section 3.1.

Proof of Proposition 3.3 (Scaling Invariance). [13, p. 12] It is fairly immediate that $X_0 = \frac{1}{a}B_0 = 0$ and since X_t just involves a time and amplitude scaling of B_t , we have that X_t is also almost surely continuous. To check independence of increments, observe that

$$X_t - X_s = \frac{1}{a}B_{a^2t} - \frac{1}{a}B_{a^2s} = \frac{1}{a}(B_{a^2t} - B_{a^2s}).$$

Since $B_{a^2t} - B_{a^2s}$ is independent of previous increments and depends only on the time difference $a^2t - a^2s$, it follows that X_t has independent increments. To check the final

property, we observe that $B_{a^2t} - B_{a^2s} \sim N(0, a^2t - a^2s)$ so $X_t - X_s$ has variance $\frac{1}{a^2}(a^2t - a^2s) = (t - s)$ and therefore, $X_t - X_s \sim N(0, t - s)$. $\square$

Before we can prove our next proposition, we should look at the covariance of Brownian motion. Suppose that $t_i \leq t_j$ and consider the covariance of B_{t_i} and B_{t_j} . We know that $\mathbb{E}[B_{t_i}] = \mathbb{E}[B_{t_j}] = 0$ and rewriting B_{t_j} as $B_{t_i} + (B_{t_j} - B_{t_i})$, we have that

$$\begin{aligned} \text{Cov}(B_{t_i}, B_{t_j}) &= \mathbb{E}[B_{t_i}B_{t_j}] - \mathbb{E}[B_{t_i}]\mathbb{E}[B_{t_j}] \\ &= \mathbb{E}[B_{t_i}(B_{t_i} + (B_{t_j} - B_{t_i}))] \\ &= \mathbb{E}[B_{t_i}^2] + \mathbb{E}[B_{t_i}]\mathbb{E}[B_{t_j} - B_{t_i}] \\ &= \mathbb{E}[B_{t_i}^2] = \text{Var}[B_{t_i}] = t_i. \end{aligned}$$

We are now ready to prove the time inversion property.

Proof of Proposition 3.4 (Time Inversion). [13, p. 13] The fact that $X_0 = 0$ is immediate by definition. It is also clear that continuity holds for every $t > 0$ so we need only check continuity at 0. Since $\mathbb{Q}$ is countable, the distribution of $\{X_t : t \in [0, \infty) \cap \mathbb{Q}\}$ is the same as for a Brownian motion, so

$$\lim_{t \rightarrow 0^+, t \in \mathbb{Q}} X(t, \omega) = 0, \text{ almost surely.}$$

Then, since $\mathbb{Q}$ is dense in $\mathbb{R}$, it follows by path continuity that

$$\lim_{t \rightarrow 0^+} X(t, \omega) = \lim_{t \rightarrow 0^+, t \in \mathbb{Q}} X(t, \omega) = 0$$

so $t \mapsto X(t, \omega)$ is almost surely continuous at $t = 0$. For the third property, we have that

$$\begin{aligned} \text{Cov}(X_{t+h}, X_t) &= \text{Cov}((t+h)B_{1/(t+h)}, tB_{1/t}) \\ &= t(t+h)\text{Cov}(B_{1/(t+h)}, B_{1/t}) \\ &= t(t+h) \cdot 1/(t+h) = t. \end{aligned}$$

Hence, $\text{Cov}(X_t, X_{t+h} - X_t) = \text{Cov}(X_t, X_{t+h}) - \text{Var}(X_t) = t - t = 0$, so X_t and $X_{t+h} - X_t$ are independent (as we have seen from ST232 that zero covariance implies independence for jointly Gaussian random variables) and thus, X_t satisfies the property of independent increments. To check the final property, we have that

$$\mathbb{E}[X_{t+h} - X_t] = \mathbb{E}[X_{t+h}] - \mathbb{E}[X_t] = 0 - 0 = 0$$

and

$$\begin{aligned} \text{Var}(X_{t+h} - X_t) &= \text{Var}(X_{t+h}) + \text{Var}(X_t) - 2\text{Cov}(X_{t+h}, X_t) \\ &= (t+h) + (t) - 2(t) = h, \end{aligned}$$

so our increments have the correct expectation and variance and thus, X_t is a standard Brownian motion. $\square$

A.2 Code

The following code was used to plot figure 1 in RStudio under the assumption that the motion of a pollen particle can be modelled by Brownian motion in two dimensions. Note that figure 1 depicts the path of a simple random walk in two dimensions with 5000 time steps so, while this visually resembles a Brownian sample path, it is important to note that this is a discrete-time process rather than a true sample path.

```
x = rnorm(5001, 0 ,1)
y = rnorm(5001, 0 ,1)
Btx = cumsum(x)
Bty = cumsum(y)
plot(Btx, Bty, type="l")
```

The following code was used to plot figure 3 in RStudio. Again, this is a discrete time process with 10000 time-steps rather than a true Brownian sample path.

```
t = seq(0,1,length = 10001)
Bt = sqrt(1/10001)*cumsum(rnorm(10001, mean = 0, sd = 1))
plot(t,Bt, type = "l", xlab = "Time", ylab = "Simulated BM")
```

The following code was used to plot Figure 4 in RStudio.

```
t.s = seq(0,1,length = 10001)[-1]
Bt = sqrt(1/10001)*cumsum(rnorm(10000, mean = 0, sd = 1))
Bt_over_t = Bt / t.s
plot(t.s, Bt_over_t, type = "l", lwd = 1,
     xlab = "Time (t)", ylab = expression(B[t]/t))
abline(h = 0, lty = 2)
```

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